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Description - This procedure ensures that the multicoil switch driver circuits on the PIN Switch Driver Board are not inadvertently damaged by transmitting with improper cable connections, and isolates faulty PIN diodes (open or shorted diodes only; this procedure may not detect leaky diodes).

Note

Record All Measurements For Troubleshooting - Record all measurements performed in the procedure for future reference as a troubleshooting aid. Refer to Table 5, Multicoil Switch Driver Circuit Data, to record the measurements.

1- TOOLS REQUIRED

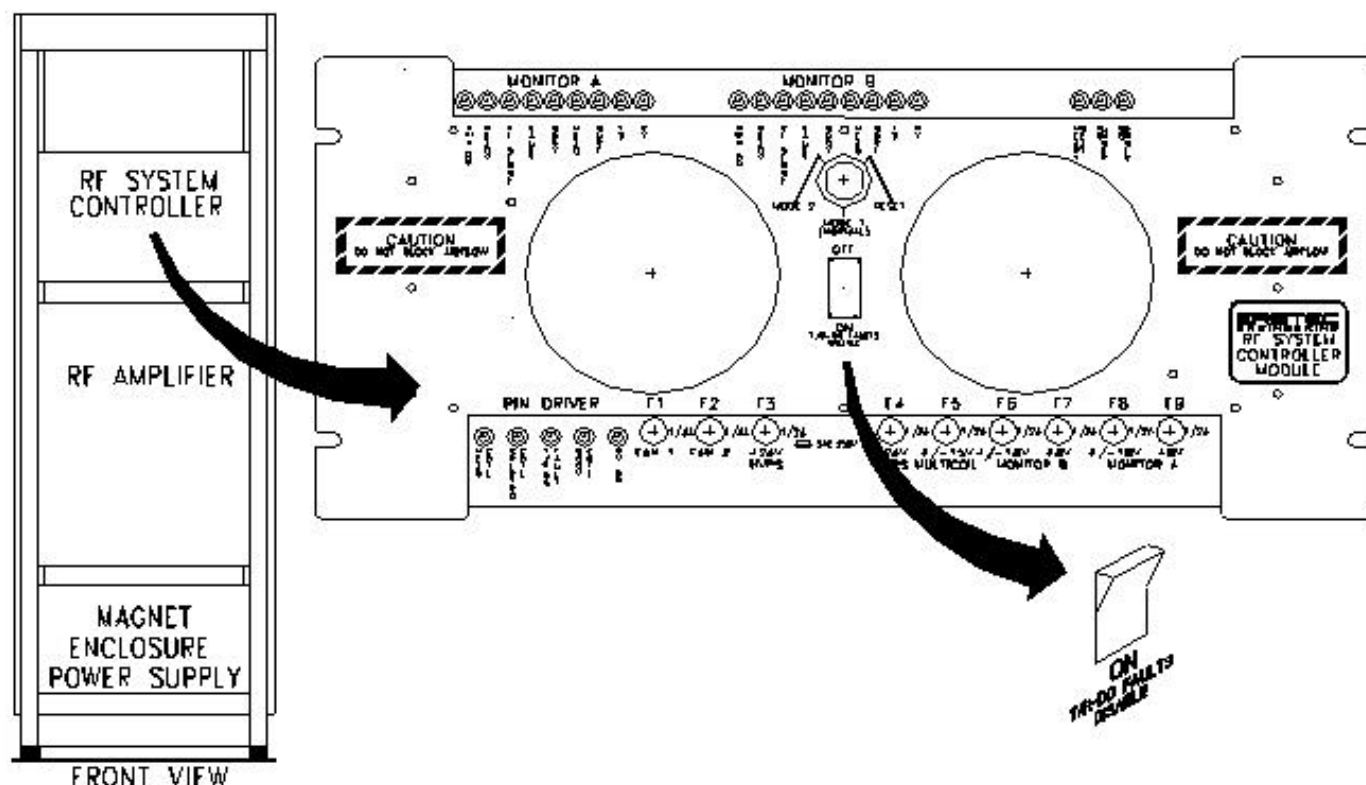
Digital multimeter, Beckman RMS 3030 or equivalent, 46-194427P49

2- MULTICOIL T/R CHECKS

Note

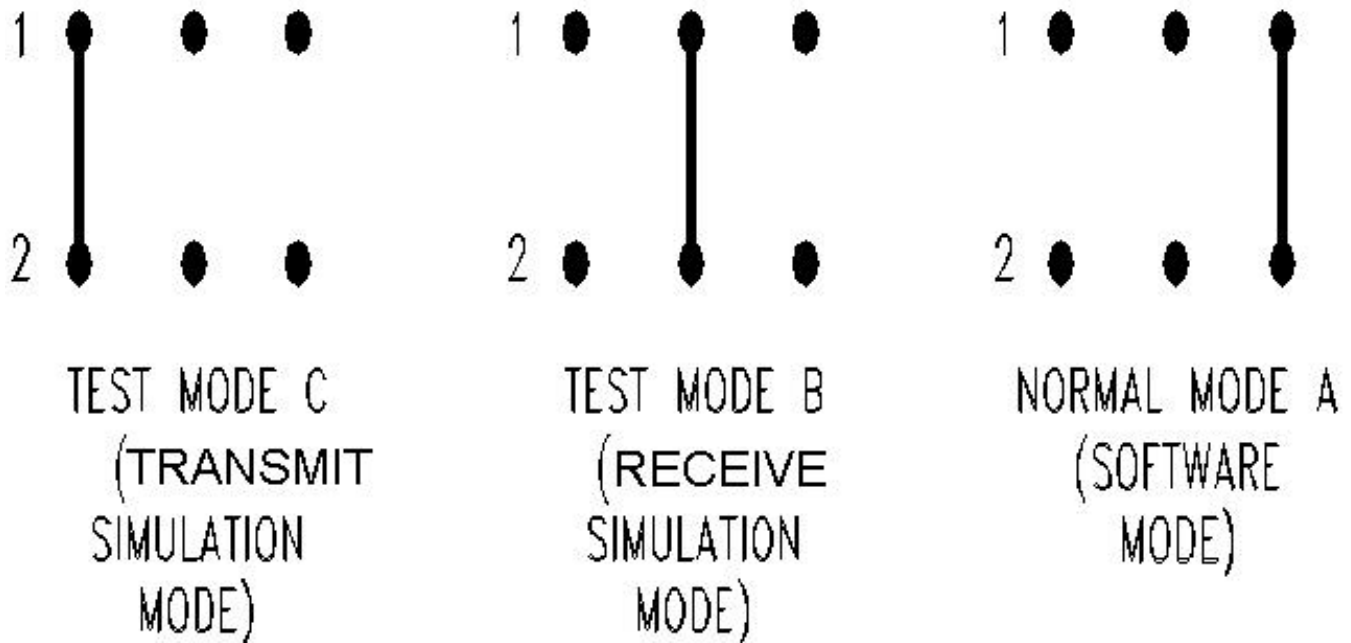
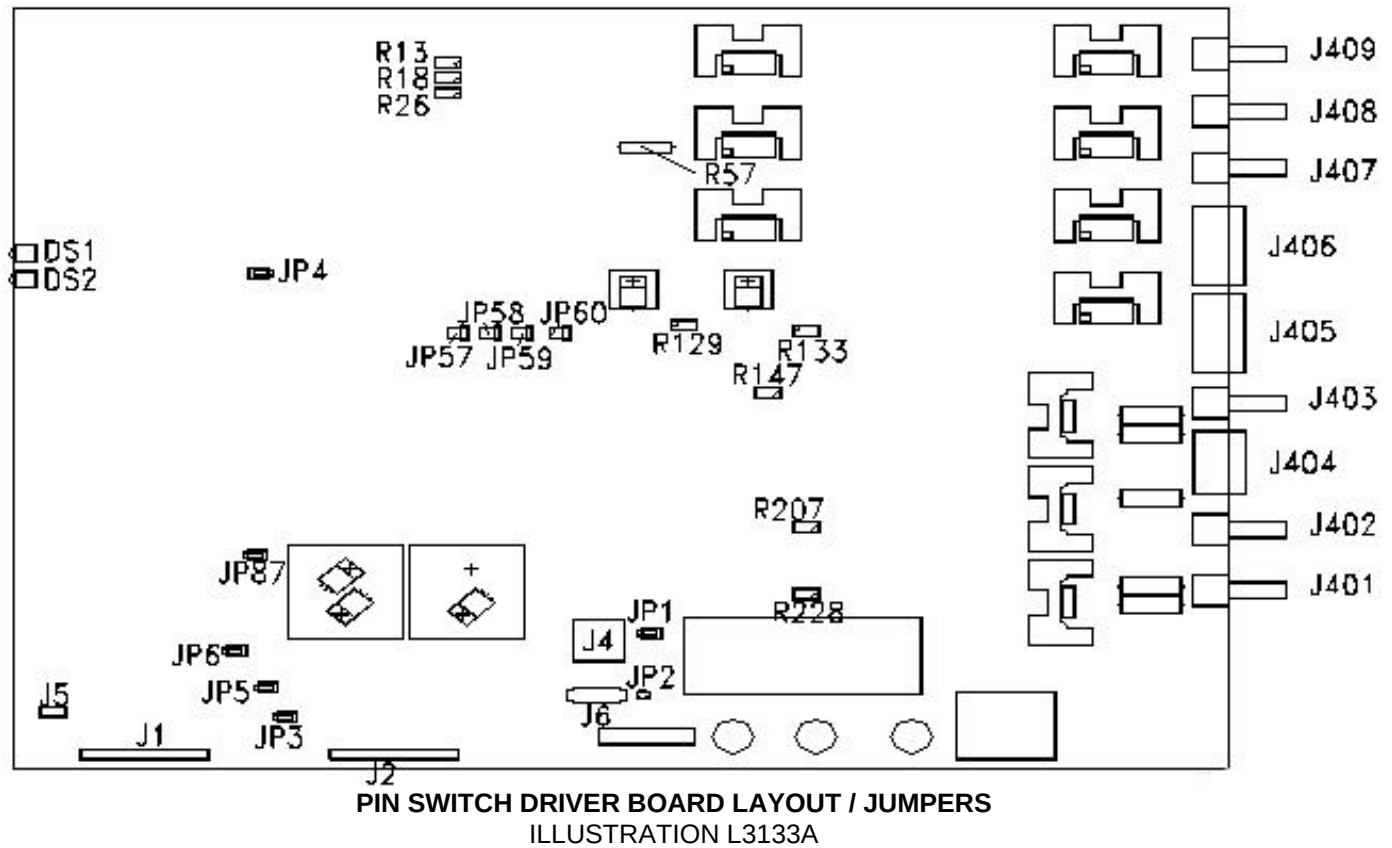
Check Coil Configuration File - Ensure that the coil configuration file values are correct.

Disable T/R Driver Fault Detection by turning on switch SW2 (T/R-DD Faults Disable on) on RF System Controller (RFSC) module. Refer to Illustration L3146a.



DYNAMIC DISABLER T/R DRIVER IN TEST POSITION
 ILLUSTRATION L3146A

- Place the jumpers on JP57, 58, 59 and 60 across pins 1 & 2 (transmit) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b.



3. Measure driver output voltages at TP2 (MCD1), TP14 (MCD2), TP20 (MCD3), and TP26 (MCD4). All voltages should be +5V $\pm$ 0.1Vdc. Refer to Table 1.

TABLE 1
MULTICOIL TR TRANSMIT CHECKS

Step	Function	Location	Specification
2-3	Driver Output Voltages	TP2,TP14, TP20, TP26	+5V $\pm$ 0.1 Vdc
2-4	Head Carriage Output Voltages (quick release connector)	B1, B3, B5, B7, B9, B11, B13, B15	+4.2V $\pm$ 0.3 Vdc
2-5	Multi-coil Select Switch DC J11 - J18	Bias Output Voltages	+14.85V $\pm$ 0.15 Vdc

Note

Measurement Reference - All measurements should be made in reference to TGND, (TP59) on the board.

4. Measure voltages at the multicoil quick release connector on the head carriage assembly, MG2A16A8 on pins B1, B3, B5, B7, B9, B11, B13, and B15 (top row, every other one). All voltages should be +4.2V $\pm$ 0.3Vdc. Refer to Table 1 above.
5. Measure voltages at the center conductors of the BNCs on the multicoil select switch, MG3A36 J11–J18. All voltages should be +14.85V $\pm$ 0.15Vdc (These voltages provide power to the preamps). Refer to Table 1 above.
6. Place the jumpers on JP5, 58, 59, and 60 across pins 3 & 4 (receive) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b above.
7. Measure driver output voltages at TP2 (MCD1), TP14 (MCD2), TP20 (MCD3), TP26 (MCD4). All voltages should be –5V $\pm$ 0.1Vdc. Refer to Table 2.

TABLE 2
MULTICOIL TR RECEIVE CHECKS

Step	Function	Location	Specification
2-7	Driver Output Voltages	TP2,TP14, TP20, TP26	–5V $\pm$ 0.1 Vdc
2-8	Head Carriage Output Voltages(quick release connector)	B1, B3, B5, B7, B9, B11, B13, B15	–4.0V $\pm$ 0.3 Vdc
2-9	Multi-coil Select Switch DC Bias Output Voltages	J11 - J18	+14.85V $\pm$ 0.15 Vdc

8. Measure voltages at the multicoil quick release connector on the head carriage assembly, MG2A16A8 on pins B1, B3, B5, B7, B9, B11, B13, B15 (top row, every other one). All voltages should be –4.0V $\pm$ 0.3Vdc. Refer to Table 2 above.
9. Measure voltages at the center conductors of the BNCs on the multicoil select switch, MG3A36 J11 - J18. All voltages should be +14.85V $\pm$ 0.15Vdc. (These voltages provide power to the preamps) Refer to Table 2 above.

Note

Voltages OK But Fault Still Exists - If the above voltages are all within tolerance, and the fault still exists, there is probably a short or open in the coil.

3- BLOWN FUSE IN RF SYSTEM CONTROLLER MODULE

1. Check the 15-pin D-sub cables connected to the multicoil filter modules (PP1A15 & PP1A16) at the penetration panel to make sure they have not been swapped.
2. Measure the cable voltages at the multicoil select switch (MG3A36J5). The voltages on MG3A36J5, pins 1, 3, and 6 should be $+15V \pm 0.2Vdc$ and pin 8 should be $-15V \pm 0.2Vdc$.
3. Measure the pin voltages at the multicoil head carriage output quick release connector (MG2A16A8). The voltages at pins A14, B14 should be $+15V \pm 0.2Vdc$ (gnd at A15), and pins A12, B12 should be $-15V \pm 0.2Vdc$ (gnd at A11).
4. Check for a bad preamp.

4- MULTICOIL DRIVER CURRENT SENSE CHECKS

1. Disconnect J20, Multicoil Select, from the rear of the RF/Pen Cabinet.
2. Place the jumpers on JP57, 58, 59, and 60 across pins 1 & 2 (transmit) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b in Section 2.
3. Measure the current sense voltages at, TP7 (MCD1), TP11 (MCD2), TP22 (MCD3), and TP28 (MCD4). See Illustration L3133b in Section 2 All voltages should be $+0.0V \pm 0.25Vdc$. Refer to Table 3.

TABLE 3
MULTICOIL TR RECEIVE CHECKS

Step	Function	Location	Specification
4-3	Unloaded (Transmit)	TP7, TP11, TP22, TP28	$0.0 \pm 0.25 Vdc$
4-5	Unloaded (Receive)	TP7, TP11, TP22, TP28	$0.0 \pm 0.25 Vdc$
4-8	M/C Chain Unloaded (Transmit)	TP7, TP11, TP22, TP28	$+2.5 \pm 0.25 Vdc$
4-10	M/C Chain Unloaded (Receive)	TP7, TP11, TP22, TP28	$-2.5 \pm 0.25 Vdc$

Note

Circuitry Detection - The circuitry detects an open circuit during transmit only, and a short during receive only. The current is sensed across a 0.69 resistor, multiplied by 7.2, and compared against reference voltages.

4. Place the jumpers on JP57, 58, 59, and 60 across pins 3 & 4 (receive) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b.
5. Measure the current sense voltages as done in step 3. See Illustration L3133b. All voltages should be $-0.0V \pm 0.25Vdc$. Refer to Table 3.
6. Reconnect J20 from the rear of the RF/Pen Cabinet. Make sure that the multicoil array is disconnected from the head carriage assembly, MG2A16A8.
7. Place the jumpers on JP57, 58, 59, and 60 across pins 1 & 2 (transmit) on the PIN Switch Driver Board. See Illustration L3133a.

8. Measure the current sense voltages as done in step 3. See Illustration L3133b in Section 2. All voltages should be $-2.5 \pm 0.25\text{Vdc}$. Refer to Table 3 above.
9. Place the jumpers on JP57, 58, 59, and 60 across pins 3 & 4 (receive) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b in Section 2.
10. Measure the current sense voltages as done in step 3. See Illustration L3133b in Section 2. All voltages should be $-2.5 \pm 0.25\text{Vdc}$. Refer to Table 3 above.
11. Connect the multicoil array to be tested to the head carriage assembly, MG2A16A8.
12. Leave the jumpers on JP57, 58, 59, and 60 across pins 3 & 4 (receive) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b in Section 2.
13. Measure the current sense voltages as done in step 3. See Illustration L3133b in Section 2. All voltages should be $-2.5\text{V} \pm 0.25\text{Vdc}$. Refer to Table 4.

TABLE 4
 MULTICOIL DRIVER CURRENT SENSE CHECKS

Step	Function	Location	Specification
5-13	M/C Chain Loaded w/ Coil(Receive)	TP7, TP11, TP22, TP28	$-2.5\text{V} \pm 0.25\text{Vdc}$
5-15	M/C Chain Loaded w/ Coil(Transmit) (6 or 8 Coil Array)	TP7, TP11, TP22, TP28	$+2.5 \pm 0.25\text{Vdc}$
	M/C Chain Loaded w/ Coil (Transmit) (4 Coil Array)	TP11, TP22	$+2.5 \pm 0.25\text{Vdc}$
		TP7, TP28	0 Vdc
	M/C Chain Loaded w/ Coil (Transmit) (2 Coil Array)	TP22	$+2.5 \pm 0.25\text{Vdc}$
		TP7, TP11, TP28	0 Vdc

14. Put JP57, 58, 59, and 60 across pins 1 & 2 (transmit) on the PIN Switch Driver Board. See Illustration L3133a & Illustration L3169b in Section 2.
15. Measure the current sense voltages as done in step 3. See Illustration L3133b in Section 2. All voltages should be $+2.5 \pm 0.25\text{Vdc}$. Refer to Table 4.

5- SYSTEM RESTORATION

1. Verify that all cables that were temporarily disconnected are restored to original configuration.
2. Turn on Fault Detection Switch SW2 on PIN Switch Driver Board (see Illustration L3146a in Section 2).

Four-channel multicoil system:

Replace the jumpers on JP57, 58, 59, and 60 to pins 5 & 6 (Software Control) on the PIN Switch Driver Board.

Two-channel multicoil system:

Nonmulticoil system:

Replace the jumpers on JP57–JP60 to pins 3 & 4 (Receive Mode). See Illustration L3133a and Illustration L3169b in Section 2.

6- MULTICOIL SWITCH DRIVER CIRCUIT DATA TABLE

TABLE 5
MULTICOIL DRIVER DATA/PIN SWITCH DRIVER BOARD

Step	Function	Location	Specification
3-3	Driver Output Voltages	TP2, TP14, TP20, TP26	+5V $\pm$ 0.2 Vdc
3-4	Head Carriage Output Voltages (quick release connector)	B1, B3, B5, B7, B9, B11, B13, B15	+4.2V $\pm$ 0.3 Vdc
3-5	Bias Output Voltages Multi-coil Select Switch DC	J11 - J18	+14.85V $\pm$ 0.15 Vdc
3-7	Driver Output Voltages	TP2, TP14, TP20, TP26	–5V $\pm$ 0.2 Vdc
3-8	Head Carriage Output Voltages(quick release connector)	B11, B13, B15 B1, B3, B5, B7, B9,	–4.0V $\pm$ 0.3 Vdc
3-9	Multi-coil Select Switch DC Bias Output Voltages	J11 - J18	+14.85V $\pm$ 0.15 Vdc
5-3	Unloaded (Transmit)	TP7, TP11, TP22, TP28	+0.0V $\pm$ 0.2 Vdc
5-5	Unloaded (Receive)	TP7, TP11, TP22, TP28	–0.0V $\pm$ 0.2 Vdc
5-8	M/C Chain Unloaded (Transmit)	TP7, TP11, TP22, TP28	+2.5V $\pm$ 0.25V
5-10	M/C Chain Unloaded (Receive)	TP7, TP11, TP22, TP28	+2.5V $\pm$ 0.25V
5-13	M/C Chain Loaded w/ Coil (Receive)	TP7, TP11, TP22, TP28	–2.5V $\pm$ 0.25V
5-15	M/C Chain Loaded w/ Coil (Transmit) (6 or 8 Coil Array)	TP7, TP11, TP22, TP28	+2.5V $\pm$ 0.25V
	M/C Chain Loaded w/ Coil (Transmit) (4 Coil Array)	TP11, TP22	+2.5V $\pm$ 0.25V
		TP7, TP28	0 Vdc
	M/C Chain Loaded w/ Coil (Transmit) (2 Coil Array)	TP22	+2.5V $\pm$ 0.25V
		TP7, TP11, TP28	0 Vdc

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 14, 1998	J. Saperstein	Initial conversion from Toolbook to Word.